

IFI 8410: Programming for Business

Syllabus for Fall 2026

Instructor: Péter Molnár

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1 Course Details

The objective of this course is to prepare you with core skills to successfully complete the MSA program. You will learn how to write programs to process and analyze data, train and evaluate machine learning models, and convey results and business values.

Day: Wednesdays
Time: 4:30 PM – 7:00 PM (Eastern Time)
Location: Classroom 501, Buckhead Center
Remote: see Zoom links on iCollege site

1.1 Instructor

Instructor: Dr. Péter Molnár
Email: pmolnar@gsu.edu
Office Phone: +1.404.413.7713
Office hours: TBA & by Appointment
Buckhead Office: Buckhead Center, 6th floor
Downtown Office: 35 Broad Street, 9th floor

During the term, it is highly recommended that you contact the instructor, in-person or via email. They are available to help you focus your projects, gain access to resources, and answer your questions. Please try to contact them at least once during the term to discuss your project. Your class members are also a good source of help.

1.2 Course Website

Class materials, assignments and due dates are posted on <https://IFI8410.molnar.ai>, Zoom links for remote access, announcements, and scores will be posted on the iCollege site.

2 Overview

This course introduces Python programming for students in data analytics and data science who have little or no prior programming experience. It starts with Jupyter notebooks for interactive exploration, then moves deliberately toward writing functions, building self-contained scripts, using the Unix command line, navigating the Unix file system, and managing code with Git. Students also encounter foundational ideas from statistics, simulation, machine learning, and algorithmic thinking through carefully scaffolded coding examples and assignments.

The course is organized around repeated practice rather than exams. Students learn by writing code every week, receiving automated feedback, revising submissions, and explaining their work in plain language. This structure supports beginners while still building habits needed for later data science, analytics, and computational coursework.

2.1 Learning objectives

By the end of the course, students should be able to:

1. Write, run, and debug Python code in Jupyter notebooks and in standalone .py programs.
2. Use variables, expressions, strings, conditionals, loops, and core Python data structures.
3. Write reusable Python functions with parameters, return values, and basic documentation.
4. Read from and write to text files and CSV files.
5. Navigate directories and files using the Unix command line.
6. Use Git to track changes, submit work, and maintain a reproducible coding workflow.
7. Apply Python to simple descriptive statistics, visualization, simulation, and introductory machine learning tasks.
8. Implement and reason about simple computational ideas such as search, hashing, shortest path, and basic data structures.
9. Interpret coding results clearly in short written explanations.
10. Use AI tools responsibly as learning aids while maintaining authorship, understanding, and academic integrity.

3 Schedule

The course schedule is shown in Table 1. However, the topics and readings may change according to the interests and abilities of the class. See the Academic Calendar. Materials may be updated 24 hours prior to class; please check before attending class.

Table 1: Course Schedule

Session	Date	Topics	Activity	Homework
1	Aug 26, 2026	Introduction to Python and the Course Environment	ACT01	-/-
2	Sep 2, 2026	Types, Strings, and Conditionals	ACT02	HW01
3	Sep 9, 2026	Loops and Core Data Structures	ACT03	HW02
4	Sep 16, 2026	Functions and Decomposition	ACT04	HW03
5	Sep 23, 2026	Files, Modules, and Scripts	ACT05	HW04
6	Sep 30, 2026	Unix File System and Command Line	ACT06	HW05
7	Oct 7, 2026	Git and Reproducible Workflow	ACT07	HW06
8	Oct 14, 2026	Python for Data Analysis	ACT08	HW07
9	Oct 21, 2026	Visualization and Descriptive Statistics	ACT09	HW08
10	Oct 28, 2026	Simulation and Randomness	ACT10	HW09
11	Nov 4, 2026	Search and Hashing	ACT11	HW10

Continued on next page

Table 1: Course Schedule (Continued)

Session	Date	Topics	Activity	Homework
12	Nov 11, 2026	Graphs and Shortest Path	ACT12	HW11
13	Nov 18, 2026	Introductory Machine Learning	ACT13	HW12
	Nov 25, 2026	<i>Thanksgiving Break</i>		
14	Dec 2, 2026	Integration and Review (last class)	-/-	-/-

Table 2: Schedule of Homework Assignments

Homework	Posting Date	Due Date
HW01	September 2, 2026	September 9, 2026
HW02	September 9, 2026	September 16, 2026
HW03	September 16, 2026	September 23, 2026
HW04	September 23, 2026	September 30, 2026
HW05	September 30, 2026	October 7, 2026
HW06	October 7, 2026	October 14, 2026
HW07	October 14, 2026	October 21, 2026
HW08	October 21, 2026	October 28, 2026
HW09	October 28, 2026	November 4, 2026
HW10	November 4, 2026	November 11, 2026
HW11	November 11, 2026	November 18, 2026
HW12	November 18, 2026	December 2, 2026

4 Required Resources

4.1 Textbook

- Wes McKinney, *Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter*, 3rd Edition, 2022, ISBN-13 978-1098104030.
Free online version at <https://wesmckinney.com/book/>.

4.2 Compute Requirements

All programming activities will be performed on the Analytics Research Cluster (ARC) using open-source software and libraries. You need a device to participate in online sessions and a device that supports web-browsers to access ARC.

Zoom capable device: you may use any device (laptop, desktop or tablet) that supports Zoom and other common video conferencing applications. Your device must be capable of transmitting your voice and video in order to actively participate in the class.

Laptop or desktop computer to complete assignments: ARC provides a web interface that supports Google Chrome, Firefox and other common web-browsers. A full keyboard and trackpad or mouse are needed to efficiently write code. The browsers on Chromebooks and iPads may not fully support the web interface.

Virtual Private Network (VPN): VPN access is needed to access ARC and other compute resources. Visit <https://technology.gsu.edu/technology-services/cybersecurity/virtual-private-network/> to configure access to the GSU-VPN on your device. (You may seek help from the Technology Service Desk.)

5 Evaluation

Students are evaluated by the deliverables summarized in the table below:

assignments	Percentage
Class Activity (top 10 of 13)	20%
Homework (top 10 of 12)	80%
Total	100%

Grading Scale:

A+	A	A-	B+	B	B-	C+	C	C-	D	F
97.0%- 100%	91.0- 96.9	89.5- 90.9	87.0- 89.4	83.0- 86.9	79.5- 82.9	77.0- 79.4	72.0- 76.9	69.5- 71.9	60.0- 69.4	Below 59.9

6 Homework Assignments

Homework assignments are a continuation of a hands-on activities completed in class. Detailed information about the activity and expectation for successful completion are provided with the instructions. See the web site for the most recent and detailed information on these assignments. **Homework assignments are individual assignments!** You may discuss the assignment with your classmates, but your final answers should reflect your individual effort. You also need to state if and how you used AI tools to help with the assignment. Completed assignments must be submitted according to the instruction given in the assignment. Due dates are posted on the class website.

7 Class Activities

Students are expected to participate in class activities during the class session. There may be more than one activity during the class session. The activities are the same for in-class and remote students. Instructions for the activity are given during the class sessions. **There are no make-up opportunities, even for excused absence.** Out of the 13 activities, the ten with the highest scores count toward the final grade.

8 Use of Internet and AI

Students are required to disclose any Artificial Intelligence (AI) assistance used in completing assignments.

The purpose of assignments is to practice what you learned and verify your understanding of concepts. Many of the assignments reflect common problems, and their solution can be found on the Internet (like <https://stackoverflow.com/>), or solved with the use of AI tools (like ChatGPT, Amazon Q, Claude Code, Codex, etc.). If you make use of these tools indicate in your program code where you found (parts of) the solution or which AI coding tool produced the section of the code.

9 Student Expectations and Class Policies

Students should plan for 2 - 3 hours of work outside of class each week for each course credit hour. Thus, a 3-credit course averages between 6 and 9 hours of student work outside of the classroom, each week. See GSU site for Academic Success: <https://success.students.gsu.edu/>

9.1 Arbitration

There will be a one-week arbitration period after graded activities are returned. Within that one-week period, you are encouraged to discuss any assumptions and/or misinterpretations that you made on the activity that may have influenced your grade.

9.2 Attendance

If you are unable to attend a class session, it is your responsibility to acquire the class notes, assignments, announcements, etc. from a classmate.

If you are enrolled in the in-person section, you may attend remotely for up to 2 sessions without approval from the instructor. Any additional remote sessions need to be approved.

If you are enrolled in the remote section, you may attend two sessions in-person without approval. **Remote participants must have their cameras on throughout the entire class period (except for breaks).** Your class participation will not be counted unless we see you on camera.

9.3 Work Progress and Submission of Deliverables

Assignments without evidence of the work-process will not be accepted. Evidence of work progress includes multiple commits to the Git repository, multiple test runs, and at least one evaluation run prior to submission. No deliverable will be accepted after the specified due date.

If you have a legitimate personal emergency (e.g., health problem) that may impair your ability to submit a deliverable on time, you must take the initiative to contact the instructor before the due date/time (or as soon after your emergency as possible) to communicate the situation.

All assignments must be submitted using the designated mechanism that is specified in the assignment. Assignments via email or any other channel will not be accepted.

9.4 Student Behavior

Behavior in class should be professional at all times. People must treat each other with dignity and respect in order for scholarship to thrive. Behaviors that are disruptive to learning will not be tolerated and may be referred to the Office of the Dean of Students for disciplinary action.

9.5 Discrimination and harassment

Discrimination and/or harassment will not be tolerated in the classroom. In most cases, discrimination and/or harassment violates Federal and State laws and/or University Policies and Regulations. Intentional discrimination and/or harassment will be referred to the Affirmative Action Office and dealt with in accordance with the appropriate rules and regulations. Unintentional discrimination and/or harassment is just as damaging to the offended party. But, it usually results from people not understanding the impact of their remarks or actions on others, or insensitivity to the feelings of others. We must all strive to work together to create a positive learning environment. This means that each individual should be sensitive to the feelings of others, and tolerant of the remarks and actions of others. If you find the remarks and actions of another individual to be offensive, please bring it to their attention. If you believe those remarks and actions constitute intentional discrimination and/or harassment, please bring it to my attention.

10 Department and University Policies

1. Prerequisites are strictly enforced. Students failing to complete any of the prerequisites with a grade of "C" or higher will be administratively withdrawn from this course with loss of tuition fees. There are no exceptions, except as granted by the instructor with the approval of the department.
2. Students are expected to attend all classes and group meetings, except when precluded by emergencies, religious holidays, or bona fide extenuating circumstances.
3. Students who, for non-academic reasons beyond their control, are unable to meet the full requirements of the course should notify the instructor, by email, as soon as this is known and prior to the class meeting. Incompletes may be given if a student has ONE AND ONLY ONE outstanding assignment.
4. A "W" grade will be assigned if a student withdraws before mid-semester if (and only if) he/she has maintained a passing grade up to the point of withdrawal. Withdrawals after the mid-semester date will result in a grade of "WF". See the GSU catalog or registrar's office for details.
5. Spirited class participation is encouraged and informed discussion in class is expected. This requires completing readings and assignments before class.
6. All exams and individual assignments are to be completed by the student alone with no help from any other person.
7. Collaboration within groups is encouraged for project work. However, collaboration between project groups will be considered cheating.
8. Copying work from the Internet without a proper reference is considered plagiarism and subject to disciplinary action as delineated in the GSU Student Handbook.
9. Any non-authorized collaboration will be considered cheating and the student(s) involved will have an Academic Dishonesty charge completed by the instructor and placed on file in the Dean's

office and the CIS Department. All instructors regardless of the type of assignment will apply this Academic Dishonesty policy equally to all students. Abstracted from GSU's Student Handbook Student Code of Conduct "Policy on Academic Honesty and Procedures for Resolving Matters of Academic Honesty" <https://codeofconduct.gsu.edu/>.

As members of the academic community, students are expected to recognize and uphold standards of intellectual and academic integrity. The University assumes as a basic and minimum standard of conduct in academic matters that students be honest and that they submit for credit only the products of their own efforts. Both the ideals of scholarship and the need for fairness require that all dishonest work be rejected as a basis for academic credit. They also require that students refrain from any and all forms of dishonorable or unethical conduct related to their academic work.

Students are expected to discuss with faculty the expectations regarding course assignments and standards of conduct. Here are some examples and definitions that clarify the standards by which academic honesty and academically honorable conduct are judged at GSU.

10.1 Wearable, Implanted, and Assistive Artificial Intelligence Devices

The use of wearable, implanted, or assistive technologies that incorporate Artificial Intelligence (AI) may be restricted during certain class activities, examinations, quizzes, laboratories, presentations, or other academic exercises. Such devices include, but are not limited to, AI-enabled eyeglasses (including prescription smart glasses), headsets, earbuds, hearing aids with AI-enhanced capabilities, wearable cameras, augmented reality (AR) or mixed reality (MR) devices, smart watches, neural interface devices, implanted technologies, and any current or future device capable of capturing, recording, transmitting, analyzing, or generating information using AI.

When the instructor designates a class activity as AI-free or device-restricted, students must be able to remove, power off, disconnect, or otherwise verifiably deactivate any wearable or implanted AI-enabled technology that could provide unauthorized assistance or capture information. Students who rely on AI-enabled prescription eyewear or similar devices may be required to use a non-AI alternative during such activities. For example, students who normally wear AI-enabled prescription glasses should maintain access to a conventional pair of prescription glasses when participating in restricted class activities.

Students who use medical, accessibility, or assistive devices containing AI capabilities must notify and obtain approval from the instructor prior to the relevant class activity whenever possible. The instructor will work with the student and, when appropriate, the University's disability or accessibility services office to determine reasonable accommodations that preserve both accessibility and academic integrity.

Failure to comply with restrictions on AI-enabled wearable or implanted devices, failure to disclose the use of such devices when required, or use of these devices to obtain unauthorized assistance may be considered a violation of course policies and academic integrity standards. The instructor reserves the right to require reasonable verification that restricted devices have been removed or disabled during designated activities.

As new technologies emerge, this policy applies to any device, whether wearable, implanted, portable, or otherwise integrated with the user, that can access, process, generate, store, record, transmit, or receive information in ways that could affect the integrity of academic work.

10.2 Plagiarism

Plagiarism is presenting another person's work as one's own. Plagiarism includes any paraphrasing or summarizing of the works of another person without acknowledgment, including the submitting of another student's work as one's own. Plagiarism frequently involves a failure to acknowledge in the text, notes, or footnotes the quotation of paragraphs, sentences, or even a few phrases written or spoken by someone else. The submission of research or completed papers or projects by someone else is plagiarism, as is the unacknowledged use of research sources gathered by someone else when that use is specifically forbidden by the faculty member. Failure to indicate the extent and nature of one's reliance on other sources is also a form of plagiarism. Any work, in whole or part, taken from the Internet or other computer-based resource without properly referencing the source (for example, the URL) is considered plagiarism. A complete reference is required in order that all parties may locate and view the original source.

The use of AI tools, including AI assistants, large language models, text generators, code generators, image generators, or similar technologies, must be disclosed and acknowledged in accordance with course and instructor requirements. Presenting AI-generated content, ideas, analyses, writing, code, images, or other work products as solely one's own, or failing to disclose the use of AI tools when disclosure is required, constitutes academic misconduct and will be treated in the same manner as plagiarism. Students are responsible for documenting and citing AI-assisted work when required and for ensuring that all submitted work accurately reflects the extent of AI involvement.

Finally, there may be forms of plagiarism or unauthorized use of AI tools that are unique to an individual discipline or course, examples of which should be provided in advance by the faculty member. The student is responsible for understanding the legitimate use of sources and technologies, the appropriate ways of acknowledging academic, scholarly, creative, or AI-assisted contributions, and the consequences of violating this responsibility.

10.3 Cheating on Examinations

Cheating on examinations involves giving or receiving unauthorized help before, during, or after an examination. Examples of unauthorized help include the use of notes, texts, or "crib sheets" during an examination (unless specifically approved by the faculty member), or sharing information with another student during an examination (unless specifically approved by the faculty member). Other examples include intentionally allowing another student to view one's own examination and collaboration before or after an examination if such collaboration is specifically forbidden by the faculty member. The use of AI tools during quizzes and exams is prohibited.

Unauthorized Collaboration. Submission for academic credit of a work product, or a part thereof, represented as its being one's own effort, which has been developed in substantial collaboration with another person or source or with a computer-based resource is a violation of academic honesty. It is also a violation of academic honesty knowingly to provide such assistance. Collaborative work specifically authorized by a faculty member is allowed.

10.4 Falsification.

It is a violation of academic honesty to misrepresent material or fabricate information in an academic exercise, assignment or proceeding (e.g., false or misleading citation of sources, the falsification of the results of experiments or of computer data, false or misleading information in an academic context in order to gain an unfair advantage).

10.5 Multiple Submissions.

It is a violation of academic honesty to submit substantial portions of the same work for credit more than once without the explicit consent of the faculty member(s) to whom the material is submitted for additional credit. In cases in which there is a natural development of research or knowledge in a sequence of courses, use of prior work may be desirable, even required; however the student is responsible for indicating in writing, as a part of such use, that the current work submitted for credit is cumulative in nature.